Yucheng Fu

(+1) 434-981-1247 zdp8uu@virginia.edu Homepage GitHub Google Scholar

Education

University of Virginia **Sep. 2024 – Jun. 2028 (Expected)**

Ph.D. in Computer Science, School of Engineering and Applied Science *Charlottesville, USA*

- Advisor: Prof. Tianhao Wang.

Sichuan University **Sep. 2020 – Jun. 2024**

B.Eng. in Cyber Security, School of Cyber Science and Engineering *Chengdu, China*

- GPA: 3.79/4.0; Rank: 1/189.

Research Interests

Key Words: Differential Privacy, Secure Multi-party Computation, Confidential LLM Systems

- The overarching goal of my research is to build practical privacy-aware systems using techniques with theoretical guarantees, e.g., differential privacy, secure multi-party computation, and oblivious algorithms on top of cryptographic protocols and trusted execution environments.
- Current research focuses: (1) designing efficient and secure LLM serving pipelines on confidential computing (TEEs); (2) secure aggregation systems with DP guarantees for various query tasks and end-to-end data synthesis.

Publications

[**C5**] Elaine Shi, Tianyao Gu, Xuanye Zheng, Yue Yang, Yiping Liu, and **Yucheng Fu**. **Multi-Party Distributed Point Functions, Revisited**. To appear in *24th Theory of Cryptography Conference (TCC)*, 2026. Randomized author order.

[**C4**] **Yucheng Fu**, Tianyao Gu, Elaine Shi, and Tianhao Wang. **Distributed Synthesis of Differentially Private Tabular Datasets**. In *35th USENIX Security Symposium (USENIX Security)*, 2026. ★ **Distinguished Paper Award Runner-Up**.

[**C3**] **Yucheng Fu**, Tianyao Gu, Elaine Shi, and Tianhao Wang. **Towards Vertically Distributed Differentially Private Synthetic Data Generation**. In *Theory and Practice of Differential Privacy Workshop Series (TPDP)*, 2025.

[**C2**] **Yucheng Fu** and Tianhao Wang. **Benchmarking Secure Sampling Protocols for Differential Privacy**. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024.

[**C1**] **Yucheng Fu**, Guoheng Sun, Han Yang, Juntian Huang, and Haizhou Wang. **Fighting Attacks on Large Character Set CAPTCHAs Using Transferable Adversarial Examples**. In *International Joint Conference on Neural Networks (IJCNN)*, 2023.

[**J1**] Guoheng Sun*, **Yucheng Fu***, Han Yang, Juntian Huang, Ruimei Zhang, and Haizhou Wang. **Enhancing the Security of Large Character Set CAPTCHAs Using Transferable Adversarial Examples**. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2025. *Equal contribution.

[**P1**] **Yucheng Fu** and Tianhao Wang. **Revisiting Continuous Noise Sampling for Multi-Party Differential Privacy**. *arXiv:2608.27766*, 2026.

Relevant Projects

Oblivious MoE Inference in Confidential Computing *Ongoing, 2026 | C++, Python, SGLang*

- Identified data-dependent expert access-pattern leakage in Mixture-of-Experts inference under CPU-GPU hybrid TEEs combining CVMs and NVIDIA Confidential Computing.
- Designed a data-oblivious framework that routes tokens to experts without leaking which experts are accessed. The implementation fuses SIMD-level permutation techniques with matrix multiplications in AVX.
- Evaluated efficiency on multiple production MoE models. For example, on DeepSeek-V4-Flash, it achieves a $16\times \sim 90\times$ speedup over existing non-oblivious solutions, and in the batched MoE model inference with various number of input/output tokens, end-to-end runtime is $0.87\times \sim 1.7\times$ compared to the non-oblivious KTransformers integrated in SGLang.

Distributed DP Synthetic Data Generation [C3, C4] *MPC, FSS, DP Synthetic Data | C++, Python, MP-SPDZ*

- Designed an MPC protocol for DP tabular data synthesis over datasets held by different organizations, achieving orders-of-magnitude speedup compared to the existing solution.
- Built, to our knowledge, the first open-source multi-party DPF implementation, which extends Google's DPF library and enables local multiplication of output shares with zero server-to-server communication.
- Received the Artifacts Available, Functional, and Reproduced badges at USENIX Security '26.

Attacking & Securing Continuous Noise Sampling in MPC [P1] *DP Attacks, MPC | Python, MP-SPDZ*

- Identified that the widely used *sample-and-scale* paradigm confines MPC-generated noise to a sparse, public fixed-point lattice, breaking the DP guarantee for any noise scale ≥ 2.
- Mounted membership-inference attacks on Orchard (OSDI '20) and DP-BREM+ (USENIX Security '25) with near-100% success under their own privacy configurations.
- Designed a provably secure fixed-point sampler with fully parallelized biased-coin flipping, cutting online rounds by up to 1,733×. The implementation is up to 706× faster than secure baselines, with utility matching the ideal mechanism.

Secure DP Noise Sampling Library & Benchmark [C2] *MPC, DP | Python, MP-SPDZ*

- Built an open-source library of 8 secure noise-sampling protocols (Laplace / Gaussian) in the MP-SPDZ framework that adds DP noise to secret-shared results with one line of code, across semi-honest and malicious MPC backends.
- Aligned all protocols to the same statistical-distance budget for a fair comparison and derived protocol-selection guidelines.

Adversarial Examples for CAPTCHA Security [C1, J1] *Adversarial ML | Python, PyTorch*

- Hardened large-character-set (e.g., Chinese) CAPTCHAs against deep-learning solvers using adversarial example techniques to mislead the black-box models used by attackers (IJCNN '23; extended in IEEE TDSC '25 as co-first author).

Honors & Awards

[2026] **Distinguished Paper Award Runner-Up**, 35th USENIX Security Symposium.
[2023] **National Scholarship** (rank 1/189 in school), Ministry of Education of China.
[2023] **The First Level Scholarship**, Sichuan University.
[2023] **Outstanding Undergraduate Graduate**, Sichuan University.
[2022] **National 1st Prize** (top 0.01% nationwide), The 8th China International College Students "Internet+" Innovation and Entrepreneurship Competition, Ministry of Education of the People's Republic of China.

Talks

USENIX Security 2026 Presentation: Distributed Synthesis of Differentially Private Tabular Datasets, Aug. 2026.
MongoDB Crypto Seminar: Distributed Tabular Data Generation with Differential Privacy, Sep. 2025.
TPDP Workshop Poster: Towards Vertically Distributed Differentially Private Synthetic Data Generation, May 2025.
CCS 2024 Presentation: Benchmarking Secure Sampling Protocols for Multi-party Differential Privacy, Oct. 2024.
UVA Security Seminar: Secure Sampling Protocols for Multi-party Differential Privacy, Sep. 2024.

Technical Skills

Programming: C/C++, Python, Java
Frameworks: PyTorch, SGLang, Opacus, MP-SPDZ, EMP-toolkit
Platforms: NVIDIA Confidential Computing, SEV-SNP, TDX
Tools: Git, Bash/Zsh, Docker, LaTeX
Languages: Mandarin (Native), English (IELTS: 7.5)